



Department of Building and Environmental Technology
Division of Energy and Building Design

Assignment 2 – Roof Analysis – Autumn 2024

VBFN15 – Moisture Safety Design of Buildings

Group 1

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1 Introduction

1 Introduction

The “VTT mould model” is a simulation tool that’s based on environmental conditions such as humidity, temperature and time of exposure to predict the risk of mould growth on layers of building materials (i.e. spruce, pine). The risk of mould growth increases gradually as the mould index increases, which is between 0-6. When the mould index is zero, it means that there is no clear mould growth, whereas an index of one sets the initial stage of mould growth. From index three on, mould is visible to the eye. [1]

Critical conditions occur, when relative humidity exceeds 80% or more (depending on the temperature), together with exposure time and temperature, which must be more than 5°C. Wood is one of the materials most susceptible to mould growth [1]. Mould can then lead to degradation of wood and other organic materials and can cause health risks. [2]

Assessing the impact of human-made climate change on buildings requires numerous scenarios to account for uncertainties. Although methods exist to systematically reduce these scenarios for faster simulations [3]. This study uses only one future scenario. Therefore, the outcome of this single study does not provide a comprehensive representation

The expected changes include increased humidity and higher temperatures, which will have a significant impact on exterior components of buildings and their cooling demand, leading to a higher risk of mould growth. [4] Thus, by 2050, many buildings designed to meet today’s standards will require modifications, such as the use of more suitable and flexible materials, controlled mechanical ventilation and improved energy efficiency and moisture management. [5],[6] This study aims to assess whether some of these potential strategies can effectively manage future weather conditions in regard to moisture safety.

2 Methodology

Utilising past climate data from Göteborg, WUFI Pro 6.8 was employed to simulate a base case for a roof design over a period of three years. The simulation settings adhered to the standard EN 13788 for indoor climate (humidity class 2). To ensure an accurate representation of boundary conditions for moisture transfer and hygrothermal performance, the air layer within the construction was divided.

The roof was oriented to the south-west with a 14-degree inclination, featuring an s_d -Value of 50.0 meters for aluminium siding and a short-wave radiation absorption rate of 0.54 for aluminium paint. Evaluations focused on water content, relative humidity (RH) and temperature (T) to assess moisture safety. Results were benchmarked against a mould growth model (parameters set for spruce, kieln-dried) using MatLab R2024a, considering critical RH and mould index values. The findings highlighted issues with the base case, presented in chapter 3.

Improvement strategies were recommended to mitigate the risks effectively. After a plausibility check in the same conditions as the base case, the roof's performance was simulated over five years using projected climate data and analysed for mould growth in the same way again. The entire process was accompanied by a risk assessment, resulting in a table shown in Appendix 1 – Risk Assessment Table.

3 Moisture Safety Assessment for a given roof

The given roof has a total thickness of 0,406 m and a U-Value of 0,111 W/m²K with the layers shown in Tab. 1. The simulation results in WUFI showed significant issues with moisture in the Layers 1, 3 and 4 with RH over 80% most of the time and a sensible, organic material [7]. However, no risk of condensation was detected in this case.

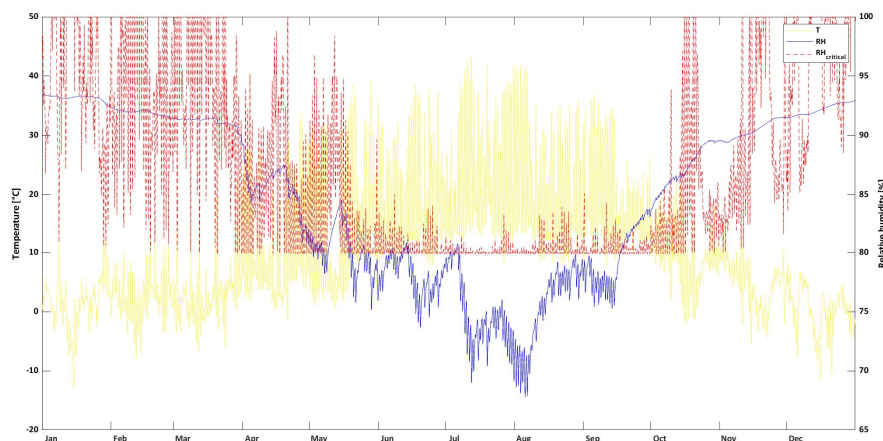
3 Moisture Safety Assessment for a given roof

Tab. 1 Setup and sketch of the roof component in the Base Case.

Nr.	s [mm]	Name	Sketch
0		Aluminium Siding ($s_d=50m$)	
1	22	Spruce radial	
2a	10	Air layer 10 mm	
2b	20	Air layer 20 mm (without additional moisture capacity)	
2c	10	Air layer 10 mm	
3	30	Wood-Fibre Insulation Board	
4	300	Mineral Wool ($\lambda=0.04$ W/mK; between wooden rafters)	
5	1	Vapour retarder ($s_d=100m$)	
6	12.5	Gypsum board	

3.1 Risk of mould growth – Layer 1 “Spruce deck”

The "Spruce deck" showed significant mould growth risk, with a maximum mould index of 1.7841. Critical points indicate times where RH and T exceeded safe thresholds for prolonged periods (Fig. 1). The first peak in the mould index occurs in December every year, being topped in the warm and rainy season in early summer right afterwards. However, a decreasing trend from year to year is visible. (Fig. 2) During dryer periods of summer the mould decreases. A clear risk of mould growth is given, though.

Fig. 1 T, RH_{crit} , RH for the wooden deck (spruce, radial) in the third year of the simulation period.

Both the mould index of the spruce deck and the wood fibre board follow a seasonal pattern over the three years of simulation - higher values during warmer months and lower during colder months. The mould index of the wood-fibre board is generally higher, which indicates that it develops more mould compared to the spruce deck. (Fig. 2)

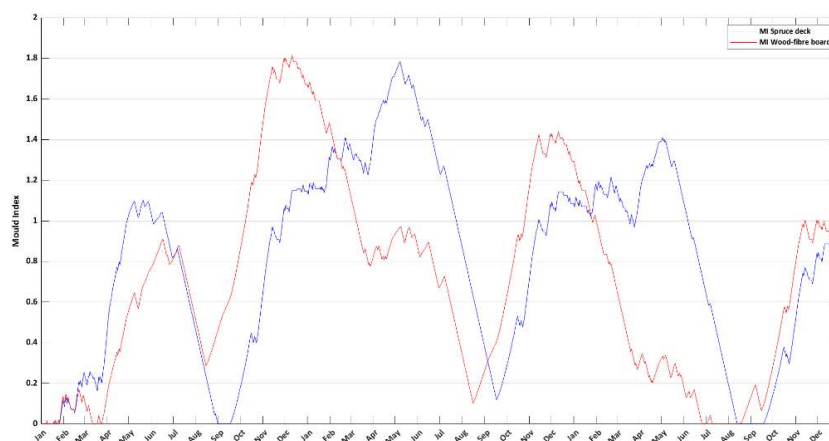


Fig. 2 Mould index for the wooden deck (spruce, radial) and the wood-fibre board over three years of simulation.

3 Moisture Safety Assessment for a given roof

3.2 Risk of mould growth – Layer 3 “Wood-fibre board”

In the wood-fibre board, conditions lead to mould growth too, as RH often tops the conditions of $RH_{critical}$ (Fig. 3), resulting in high mould index values. The upper limit is 1.8148, which means that it has not reached the critical limit of visibility yet. (Fig. 2). However, after the first year in this layer, the mould index begins to decrease with a linear trend, dropping below one in the third year.

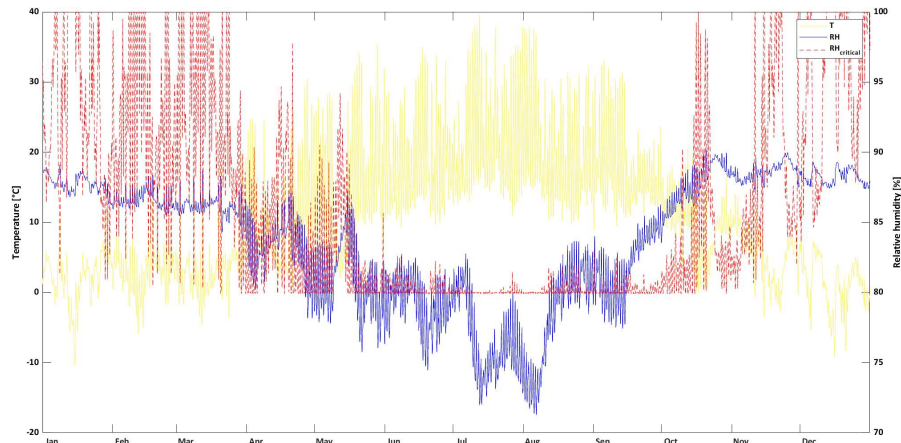


Fig. 3 T, RH_{crit} , RH for the wood-fibre insulation board in the third year of the simulation period.

3.3 Risk of mould growth – Layer 4 “Mineral wool insulation”

Although there is no risk of condensation in the insulation layer, RH rises to quite high levels reaching more than 80% from time to time. Throughout the entire layer, temperatures also reach 20°C and more. The worst conditions appear in the outermost layer.

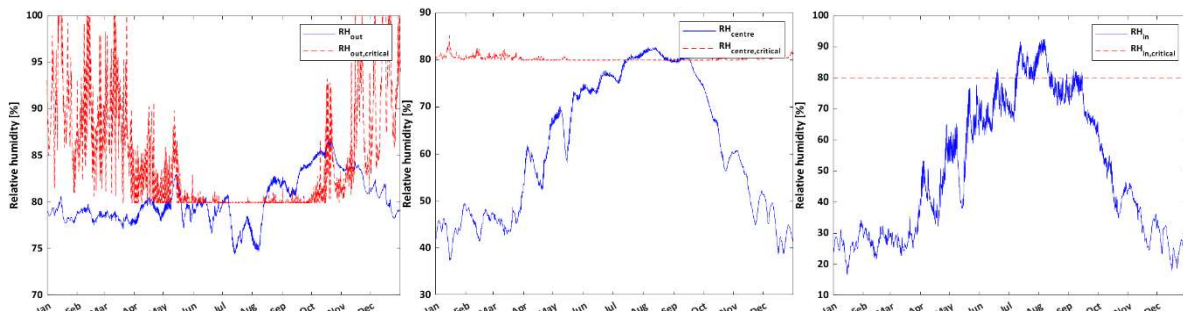


Fig. 4 RH_{crit} , RH for three different positions in the insulation layer in the third year of the simulation period. Outside = 0,0032m, Centre = 0,1501m, Inside = 0,2989m.

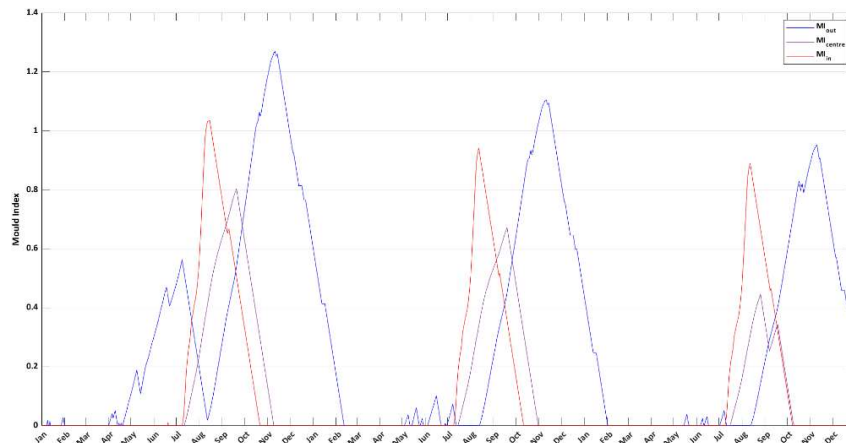


Fig. 5 Mould index for three different positions in the insulation layer over three years of simulation. Outside = 0,0032m, Centre = 0,1501m, Inside = 0,2989m.

3 Moisture Safety Assessment for a given roof

This is enough reason to closely look at the mould growth risk in the mineral wool between wooden rafters. Fig. 4 and Fig. 5 amplify the suspicion that the outside of the insulation shows the most risk for mould growth reaching the highest mould growth index of 1,2688. In general, it shows a decreasing trend over time, with its maximum in the first year. Furthermore, a time delay in mould growth is seen from inside to outside in the insulation, due to the duration of diffusion and heat transfer processes.

3.4 Summary of risk

Over three years, the mould growth assessment for the base case, considering initial conditions, shows significant moisture exposure in critical layers. The spruce deck as well as the wood-fibre board show the most favourable conditions for mould growth. Meanwhile, the mineral wool (outside) displays a lower mould index but still shows notable moisture presence. (Tab. 2)

Tab. 2 Summary of Mould Growth Assessment for the sensible layers. Base case, three years, incl. initial conditions.

Layer	RH _{max} [%]	RH _{mean} [%]	T _{mean} [°C]	MI _{max}	MI _{mean}	MGR <0 [h-%]
Spruce deck	93.97	85.39	9.95	1.78	0.79	47.21
Wood-Fibre insulation	92.34	84.44	10.55	1.81	0.70	44.20
Mineral Wool (outside)	90.69	81.16	11.48	1.27	0.30	59.73

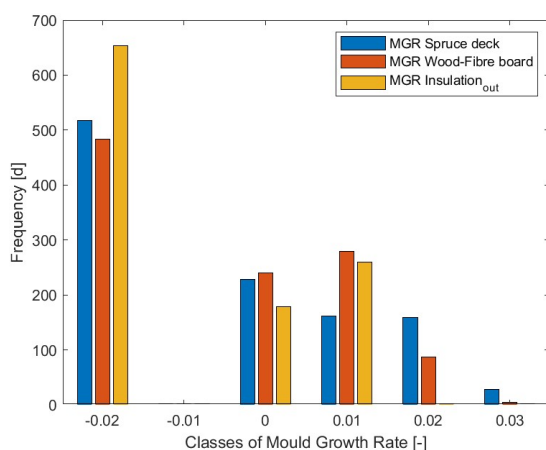


Fig. 6 Count of Mould Growth rate per class and per layer over three years of simulation for the base case. Borders: -0.02, -0.01, 0, 0.01, 0.02, 0.03, 0.04

Fig. 6 illustrates the distribution of mould growth rates over three years of simulation for all relevant layers. It is evident that on many days, a decrease in mould occurs. Additionally, it shows that the spruce layer has the highest range of mould growth rates. The most stable and most safe conditions occur in the insulation layer.

These findings indicate a potential for compromised structural integrity, reduced insulation performance, and potential health risks due to mould. Effective moisture management and mitigation strategies are essential to prevent these adverse effects.

3.5 Measurements obtained

Air exchange rates in ventilated cavities of roofs occur in a range of 11/h to 84/h [8]. No major changes were observed within this range, except at 10/h where the results showed more improvement. A representative value of 25/h was chosen. Other measurements are not as effective. However, to prevent the moisture carried in this air flow to spread into the insulation layer and for the practical reason of having a second waterproof layer (back up sealing), a weather barrier ($s_d=0,1m$) was applied on top of a now less hygroscopic wood fibre board (high density with insulation effect).

The thickness of the mineral wool was decreased by two centimetres to adapt to the U-value of the given reference. Furthermore, an OSB 3 plank was set below the insulation layer for practical reasons and has a rather high s_d -value. Although changes in the vapour barrier ($s_d=10-1500m$) did not show any significant impact to the water content, the material of the base case is still applied.

4 Moisture Safety Assessment for an improved roof in future climate

3.5.1 Plausibility Check

Applying the proposed mould model on the improved version with the climatic conditions of the base case as a plausibility test leads to great improvements. Across all layers only one event of mould risk happens, the Spruce deck basically does not show mould growth (max MI = 0.0442). The maximum mould index is 0.5589 in the Rigid Underlay board and 1.4496 in the insulation layer.

4 Moisture Safety Assessment for an improved roof in future climate

The improved roof has a total thickness of 0,407 m and a U-Value of 0,115 W/m²K with the layers shown in Tab. 3. The simulation results showed significant issues with moisture in the Layers 1, 4 and 5 [7]. The added layer of the OSB did not show any risk. No risk of condensation was detected in this case.

Tab. 3 Setup and sketch of the roof component in the improved version.

Nr.	s [mm]	Name	Sketch
0		Aluminium Siding ($s_d=50m$)	
1	22	Spruce radial	
2a	10	Air layer 10 mm	
2b	20	Air layer 20 mm (without additional moisture capacity)	
2c	10	Air layer 10 mm	
3	1	TRASPIR EVO 160 ($s_d=0,1m$)	
4	35	AiF Rigid Underlay Board WF high density	
5	280	Mineral Wool ($\lambda=0.04$ W/mK; between wooden rafters)	
6	15	AiF OSB 3	
7	1	Vapour retarder ($s_d=100m$)	
8	12.5	Gypsum board	

4.1 Analysis of changes in used weather data

Another step taken was using one out of many possible future weather scenarios for the location of Göteborg in a separate simulation of the improved roof component on moisture transfer over a five-year period. Hence the results we discuss in the following do not only talk about a moisture safe design in the past climate, but they also consider one possibility of different future conditions, caused by human made climate change, which are mostly considered as more challenging for buildings.

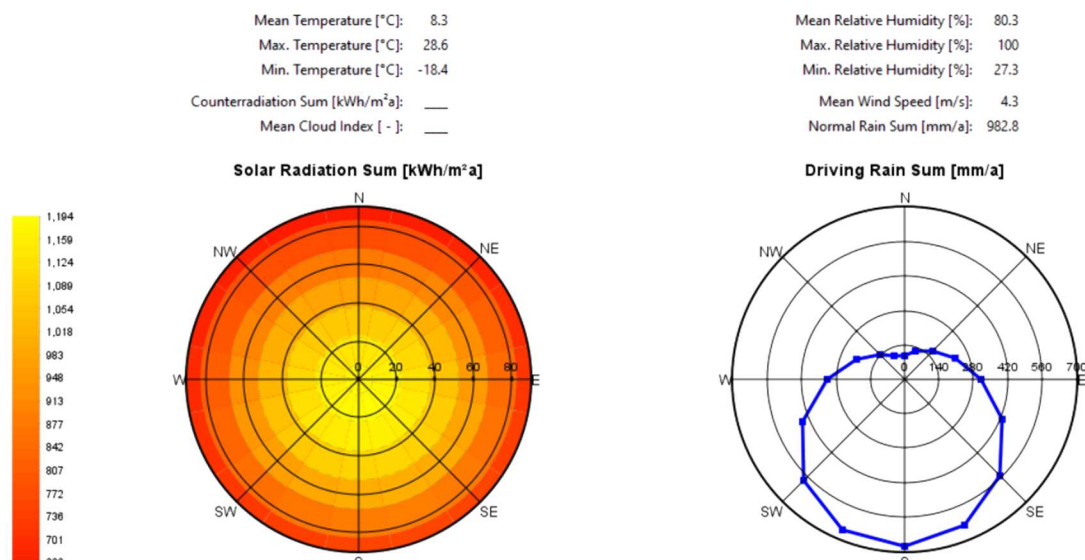


Fig. 7 Climate analysis of the future weather file.[9]

4 Moisture Safety Assessment for an improved roof in future climate

The main changes to the base case in the weather file are a higher RH of the outdoor air (mean RH rises by 6.3%) and less rain in general but more wind driven rain (up to 700 mm/a instead of 550 mm/a). Moreover, the mean temperature decreases by 0.5°C, although the amplitude of it is way higher compared to the typical meteorological year of the Base Case – extreme weather events are considerably more likely in the future. These changes have a negative impact on the moisture safety of our component. On the other hand, a significant increase in solar radiation is a positive aspect. (Fig. 7)

4.2 Risk of mould growth – Layer 1 “Spruce deck” and Layer 3 “Rigid Underlay board”

For almost the entire observation period RH remains below $RH_{critical}$, indicating that conditions have been successfully managed to prevent mould development. However, there is an intersection between RH and $RH_{critical}$ at the initial parts of the graph. This could be due to the initial moisture inside the materials or the climatic conditions outside that may cause temporary humidity variations. (Fig. 8)

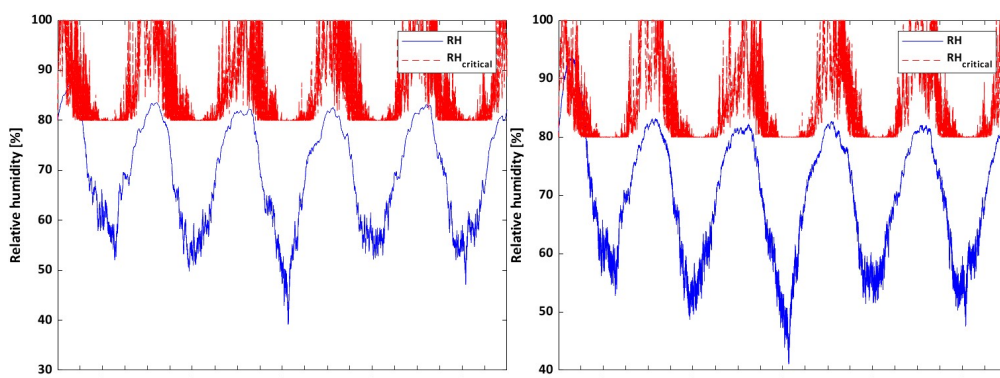


Fig. 8 RH_{crit} and RH in the Spruce deck and the Rigid Underlay board over five years of simulation in improved version.

The mould index remains close to zero throughout most of the five-year period, with only a brief spike observed early in the timeline. This indicates that the conditions were generally not conducive to mould formation, except for a short period where favourable conditions allowed minor mould growth. The consistently low mould index demonstrates effective moisture management, ensuring mould did not take hold over extended periods. (Fig. 9)

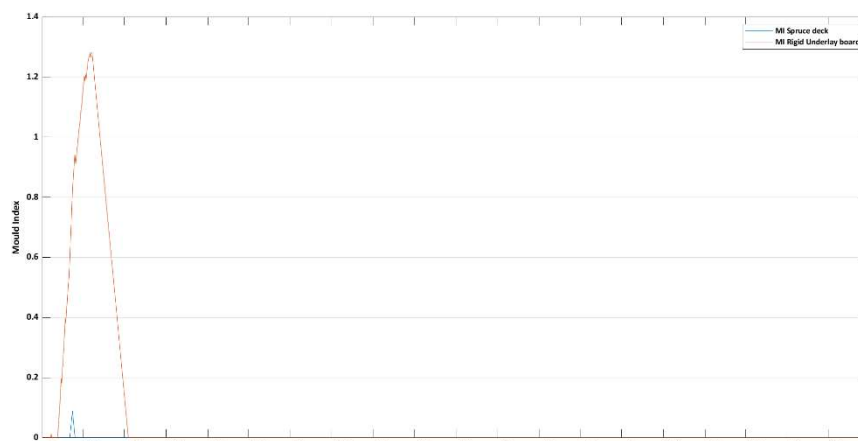


Fig. 9 Mould index over five years of simulation in the Spruce deck and Rigid Underlay board in the improved version.

4.3 Risk of mould growth – Layer 5 “Mineral wool insulation”

Apart from the initial conditions the moisture load in the insulation layer does not allow mould growth. Fig. 10 (left) shows the critical RH and RH for the outermost position in the mineral wool, which are only close to each other in the beginning. For further information RH is also plotted for the other positions, staying below 70% all the time.

4 Moisture Safety Assessment for an improved roof in future climate

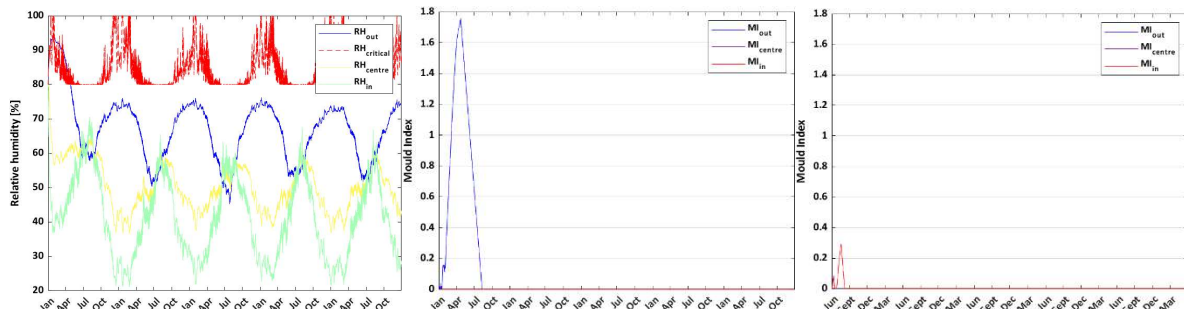


Fig. 10 RH_{crit} for the outermost position; RH (left) and mould index (middle) for three different positions in the insulation layer over five years of simulation. Mould index for simulation start in June (right).

Outside = 0,0032m, Centre = 0,1501m, Inside = 0,2989m. Aid lines for 20°C and 80% RH.

The importance of the initial conditions becomes more evident when comparing the mould index for a simulation that starts in January (winter) to a simulation that starts in July (summer). With summer conditions promoting drying out no mould occurs. (Fig. 10, middle, right). Otherwise, the mould index reaches 1.7586 in its peak in June.

4.4 Summary of risk

Over five years, the mould growth assessment of various critical layers in future climate demonstrated notable improvements in moisture management and mitigation. For the spruce deck values indicated very low mould growth. The rigid underlay board displayed slightly higher mould indices but still effective moisture control. Mineral wool (outside) showed higher RH and T but maintained a relatively low mould growth indices, especially when conquering bad initial conditions by starting the simulation in summer months. (Tab. 4) The distribution of mould growth rates enhances this conclusion with barely any high positive rates. (0).

Tab. 4 Summary of Mould Growth Assessment for the sensible layers. Improved version, five years, incl. initial conditions.

Layer	RH _{max} [%]	RH _{mean} [%]	T _{mean} [°C]	MI _{max}	MI _{mean}	MGR <0 [h-%]
Spruce deck	87.81	69.49	10.35	0.09	0.00	89.05
Rigid Underlay WF board	93.58	70.27	10.87	1.28	0.06	89.65
Mineral Wool (outside)	94.31	67.17	11.85	1.76	0.10	92.88
OSB 3	80.00	42.44	20.67	0.00	0.00	99.89

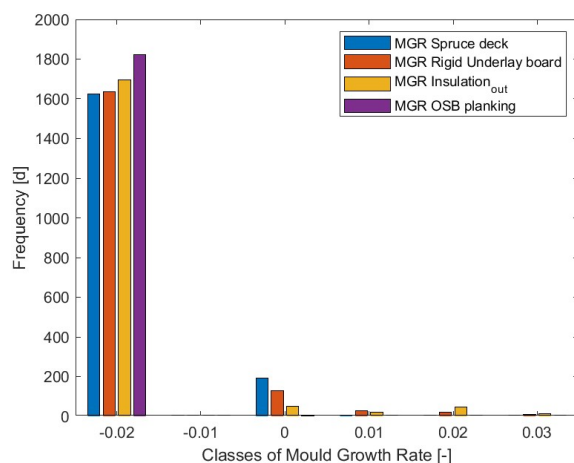


Fig. 11 Count of Mould Growth Index per class and per layer over five years of simulation for the improved version. Borders: -0.02, -0.01, 0, 0.01, 0.02, 0.03, 0.04.

The findings indicate significant reductions in mould growth indices, more controlled relative humidity levels, and overall improvement in moisture management, underscoring the effectiveness of the design improvements in mitigating mould risks.

In Fig. 12 the water content of all the analysed layers is shown over time. It fluctuates seasonally, with higher values during warmer months and lower levels in colder periods for the outer layers. Within the layer of the OSB seasonal changes work the other way around.

5 Discussion

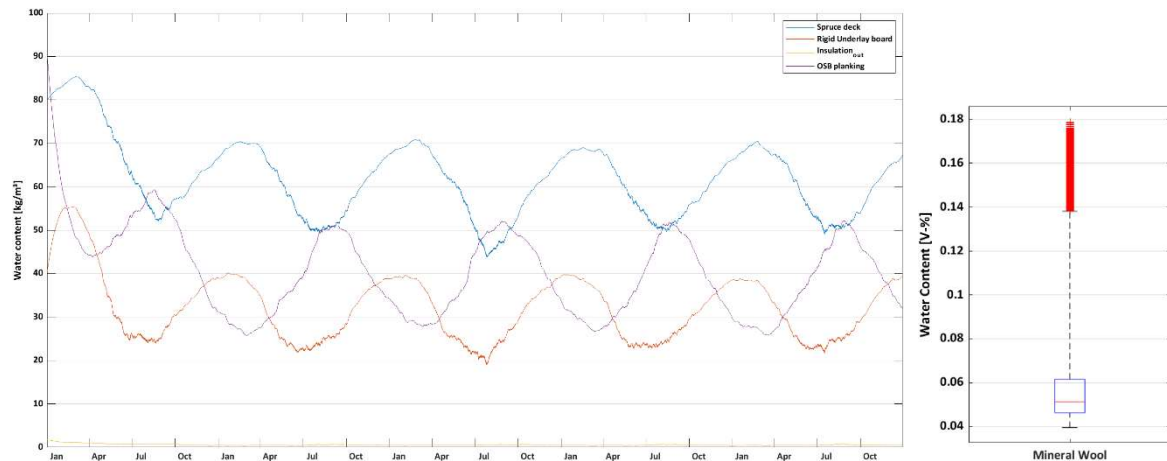


Fig. 12 Left: Water content of all sensible layers over five years of simulation for the improved version. Right: Boxplot of the water content in the insulation layer in V-%, top whisker 5x 25-75th percentile.

These fluctuations indicate the impact of changes in (especially exterior) moisture and heat conditions throughout the whole roof component due to the slow diffusion and heat transfer processes. Already after the first cycle the conditions remain very stable, an equilibrium is reached quickly. Else than the wooden products, the mineral wool cannot store a lot of moisture and therefore shows almost no water content. Still the materials heat conductivity is very sensible to changes in moisture content in a negative way, the boxplot on the right side of Fig. 12 shows the very low rate of moisture content which does not yet affect the mineral wools thermal properties to a bigger extend. [10]

5 Discussion

5.1 Measurements taken

The changes and modifications obtained in chapter 3.5 led to good results regarding the improved version of the roof. However, the most beneficial factor for a moisture safe design is the ventilation of the air cavity. Because of this change it is mandatory to have a weather barrier on top of the insulation layer. A stable rate for ventilation cannot be guaranteed as wind, snow, ice and maintenance of the roof can influence it. Over a range of 10 to more than 40 air changes per hour the moisture safety was still proven through more simulations.

The mould index has decreased slightly from 1.81 to 1.76, showing the improvements' effectiveness. Notably, the share of mould growth rates has shifted towards negative values, indicating reduced mould development. Overall, the comparison highlights a notable improvement in moisture control, reducing mould risk across the structure and improving durability under future weather conditions.

5.2 Moisture safety for building constructions in future climate scenarios

Due to a lack of information on which climate scenario is used as a base for the weather file with future weather data no resilient arguments can be stated. However, Scandinavian regions will experience a drastically higher impact on weather by global warming then the global mean [11]. With -0,5°C our weather data seems to show the result of a very-low to negative-emission scenario. Because of a higher amplitude we can sense that the higher probability of extreme weather events is taken into account.

As climate research claims for many years, a change is not only seen in temperature but also in precipitation. Even though it is hard to model, worsening conditions regarding moisture impact on buildings are evident to be an issue. The used weather scenario shows a mix of various changes and is not seen as a catastrophically bad one. The assessed roof component together with the suggested measurements however shows good resilience for this future weather when it comes to a moisture safety design.

Appendix 1 – Risk Assessment Table

Appendix 1 – Risk Assessment Table

Tab. 5 Risk assessment table for a moisture safe building construction. Includes "ByggaF" [12].

Phase	Risk	Impact	Probability	Consequence	Rating	Mitigation
Material Quality	Use of substandard materials	Reduced durability, potential leaks, different properties	3	3	9	Source high-quality materials, conduct regular quality checks, material properties in contracts
Installation Errors	Incorrect layering, alignment, construction damages	Structural weakness and water infiltration	3	3	9	Hire experienced professionals, provide thorough training
Weather Conditions	Adverse weather conditions	Delays and reduced work quality	3	2	6	Monitor weather forecasts, plan around favorable conditions, provide weather-proof construction site, focus on pre-manufacturing
Maintenance Issues	Neglecting routine maintenance	Accelerated wear and tear, higher long-term costs	3	2	6	Establish a regular maintenance schedule, conduct timely repairs, follow "ByggaF" guidelines

Probability	Consequence				Risk Level	
	Low	1	2	3	4	1 to 4 Low Risk
	Medium	2	4	6	8	5 to 8 Medium Risk
	High	3	6	9	12	9 to 12 High Risk
	Very High	4	8	12	16	13 to 16 Very High Risk

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